

Cultures

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```
library(tidyverse)
library(readxl)
library(ggplot2)
```

Reading in Data

```
#read in data
tib <- read_excel('C:/Users/Eddie/OneDrive/Desktop/BIOSTAT/Thesis/Thesis/DataProcessed/Cultures.xlsx', sheet = "tissue")
ggo <- read_excel('C:/Users/Eddie/OneDrive/Desktop/BIOSTAT/Thesis/Thesis/DataProcessed/Cultures.xlsx', sheet = "ggo")
cons <- read_excel('C:/Users/Eddie/OneDrive/Desktop/BIOSTAT/Thesis/Thesis/DataProcessed/Cultures.xlsx', sheet = "consolidated")
bronch <- read_excel('C:/Users/Eddie/OneDrive/Desktop/BIOSTAT/Thesis/Thesis/DataProcessed/Cultures.xlsx', sheet = "bronch")
atel <- read_excel('C:/Users/Eddie/OneDrive/Desktop/BIOSTAT/Thesis/Thesis/DataProcessed/Cultures.xlsx', sheet = "ateloid")
ln <- read_excel('C:/Users/Eddie/OneDrive/Desktop/BIOSTAT/Thesis/Thesis/DataProcessed/Cultures.xlsx', sheet = "lymph node")
thin <- read_excel('C:/Users/Eddie/OneDrive/Desktop/BIOSTAT/Thesis/Thesis/DataProcessed/Cultures.xlsx', sheet = "thin")
thick <- read_excel('C:/Users/Eddie/OneDrive/Desktop/BIOSTAT/Thesis/Thesis/DataProcessed/Cultures.xlsx', sheet = "thick")
```

Visualize the Distribution of Scores

Tree-in-bud

```
score_colors_new <- c(
  "3" = "#E34A33", # Orange-Red for Score 3
  "2" = "#3B9142", # Green for Score 2
  "1" = "#5EC9C2", # Light Blue for Score 1
  "0" = "white"   # white for Score 0
)

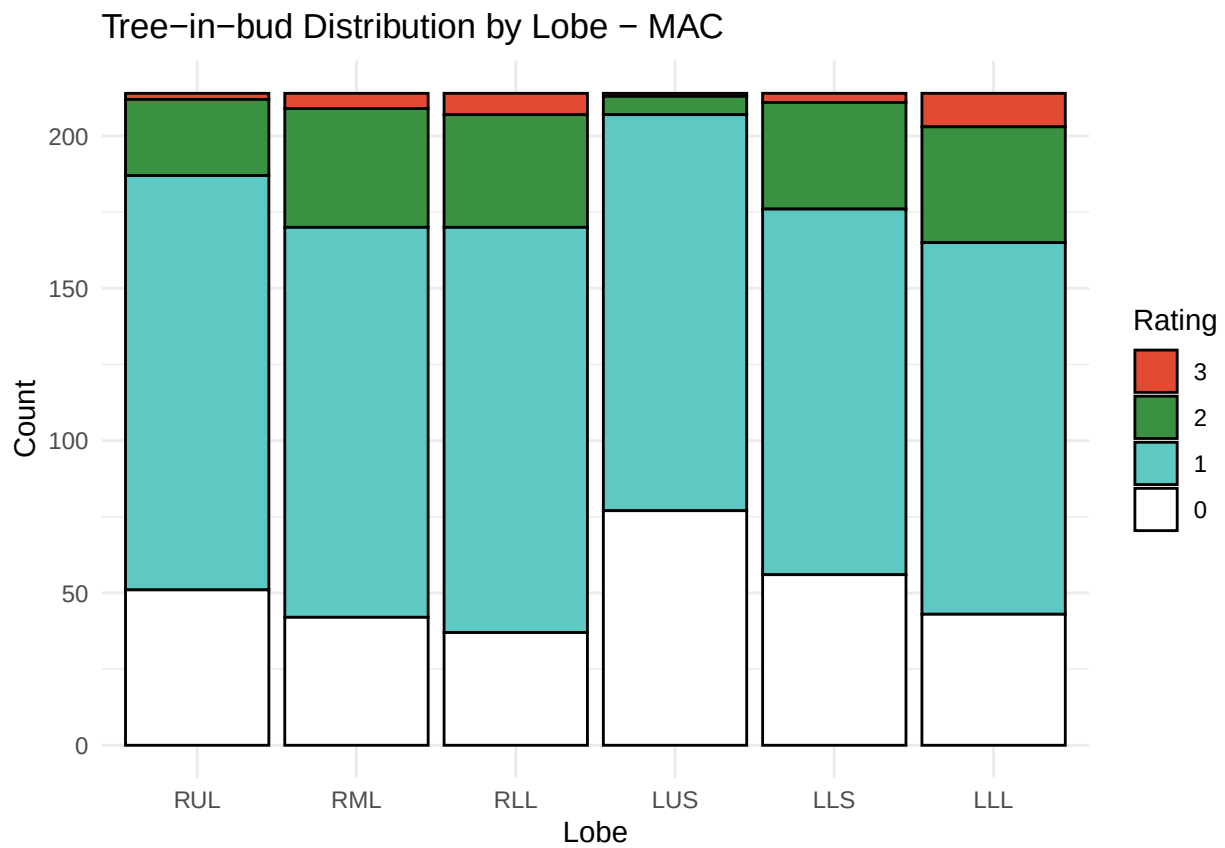
plots <- tib %>%
  mutate(
    score = factor(score, levels = c(3,2,1,0)),
    score_color = score_colors_new[score]
```

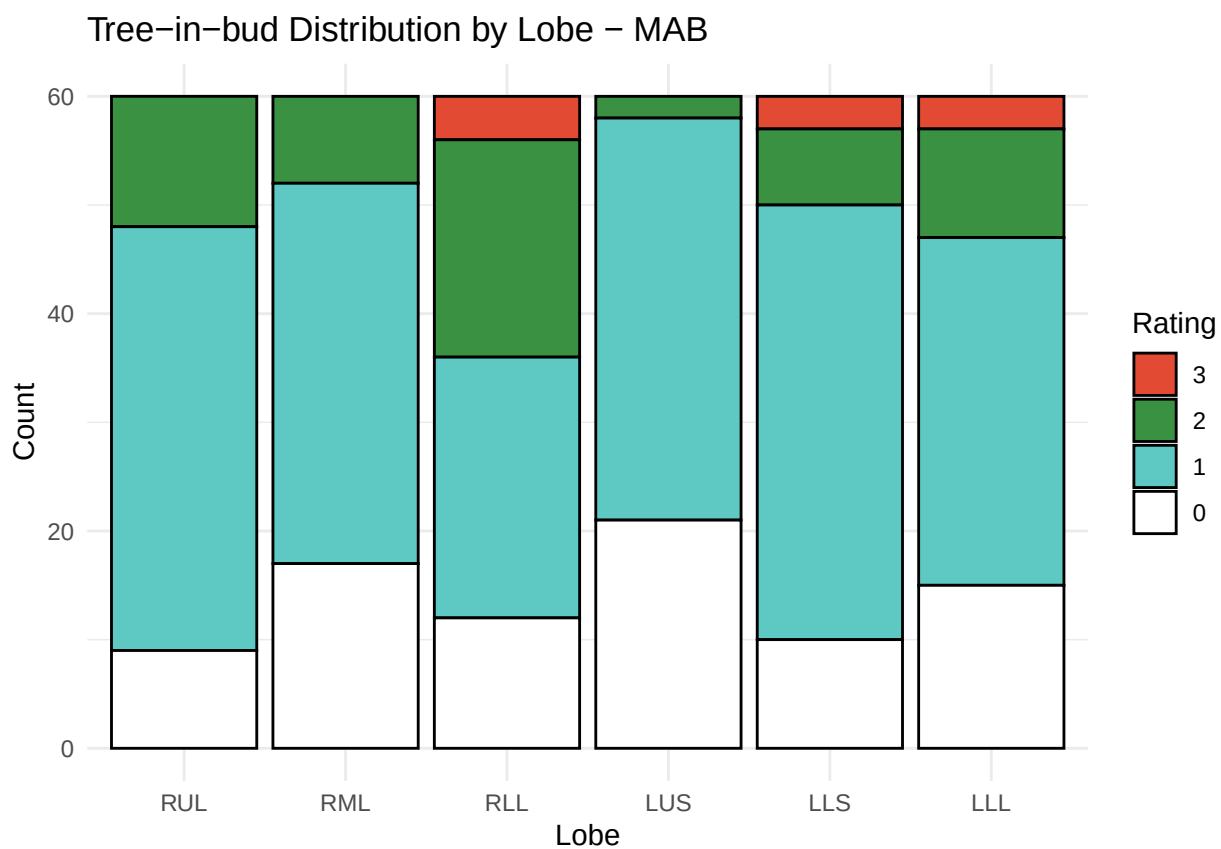
```

lobe = factor(lobe, levels = c("RUL", "RML", "RLL", "LUS", "LLS", "LLL")),
pathcode = factor(pathcode, levels = 1:4,
                  labels = c("MAC", "MAB", "MAC + MAB", "Other"))
) %>%
split(.$pathcode) %>%
lapply(function(df) {
  ggplot(df, aes(x = lobe, fill = score)) +
    geom_bar(position = "stack", color = "black") +
    scale_fill_manual(values = score_colors_new, name = "Rating") +
    labs(title = paste("Tree-in-bud Distribution by Lobe -", unique(df$pathcode)),
         x = "Lobe", y = "Count") +
    theme_minimal()
})

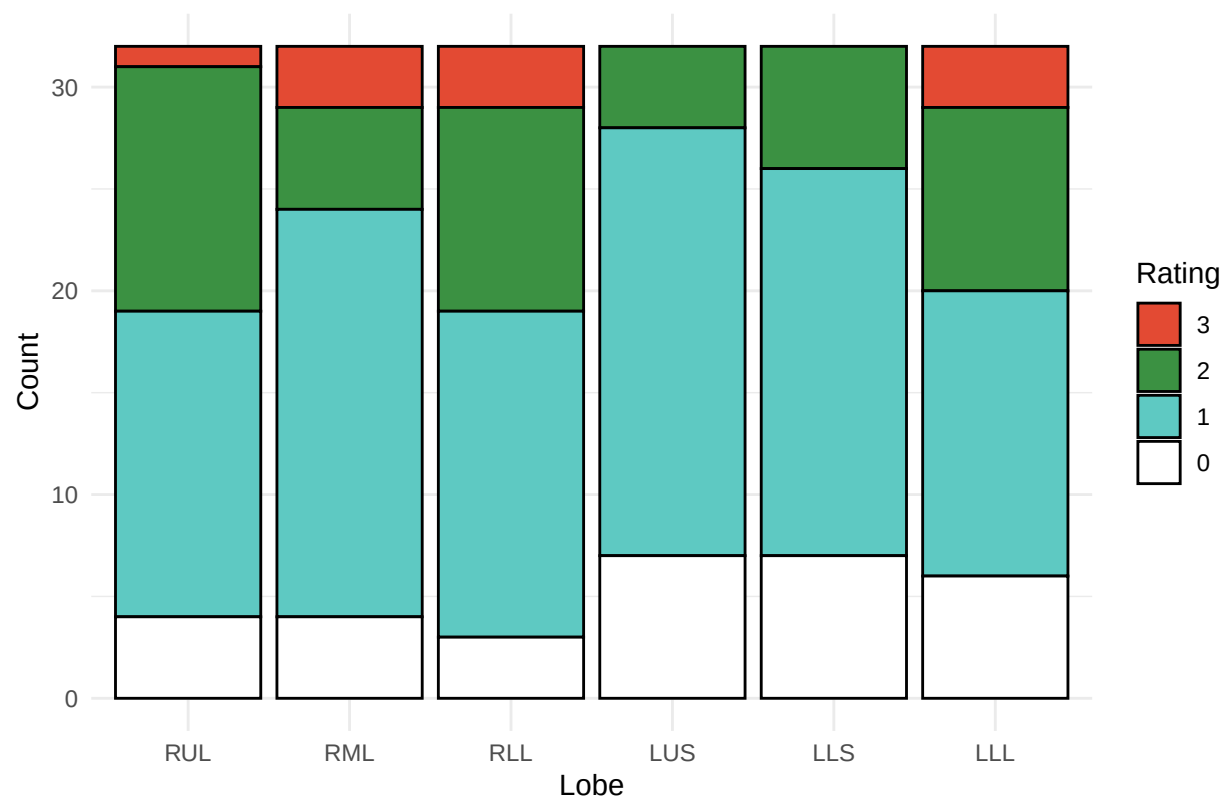
# Print
plots[[1]]; plots[[2]]; plots[[3]]; plots[[4]]

```

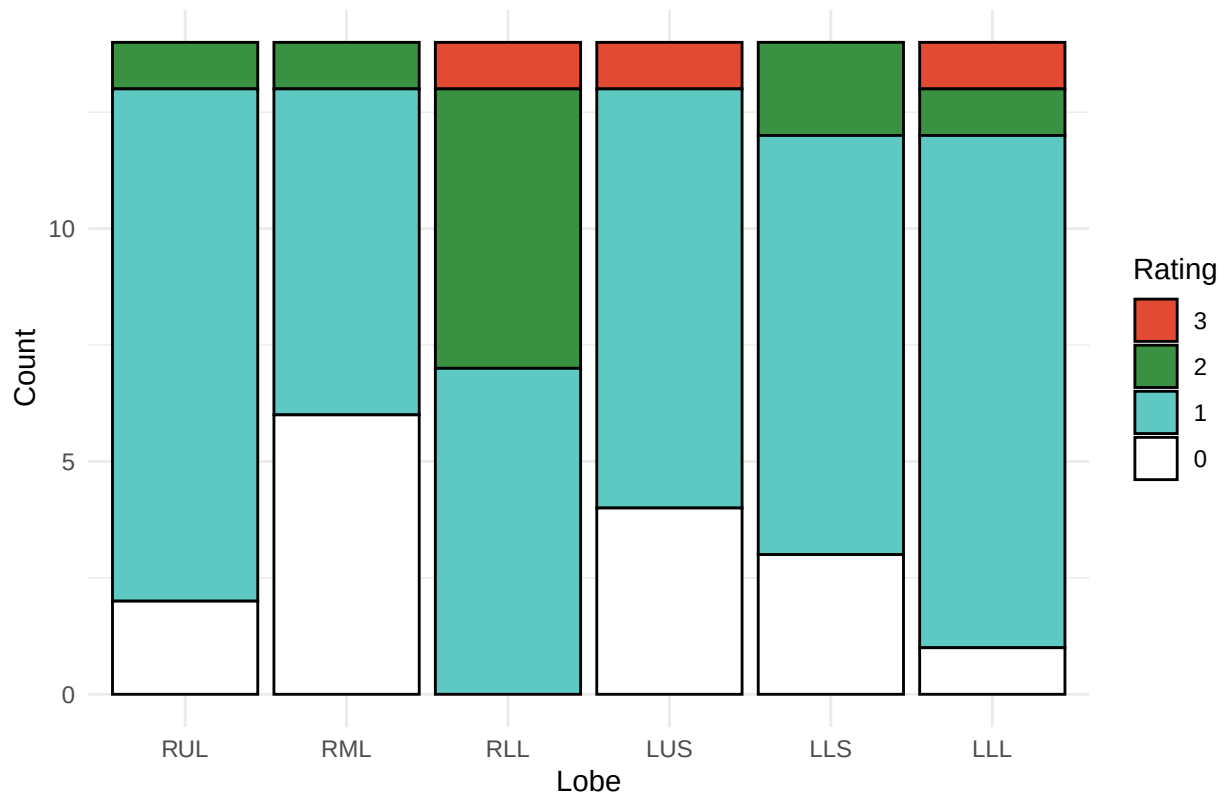




Tree-in-bud Distribution by Lobe – MAC + MAB



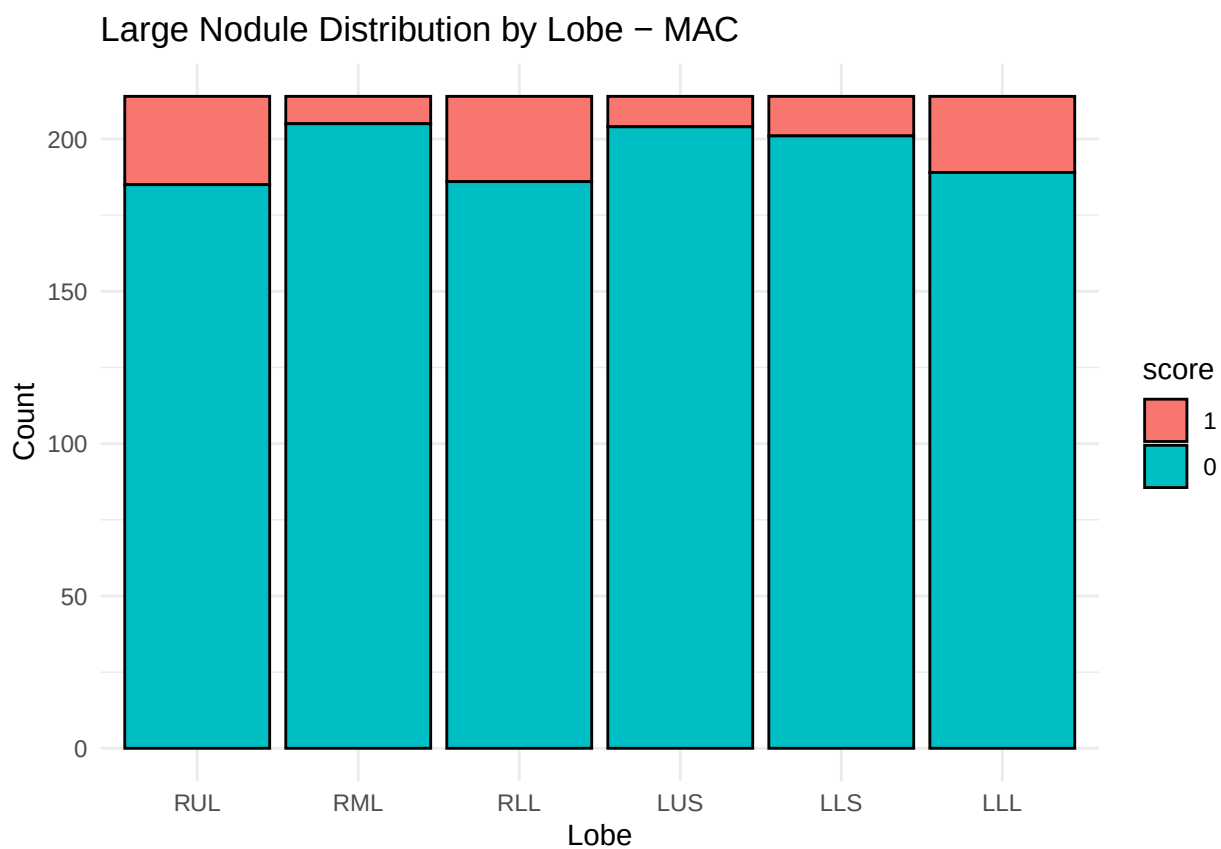
Tree-in-bud Distribution by Lobe – Other

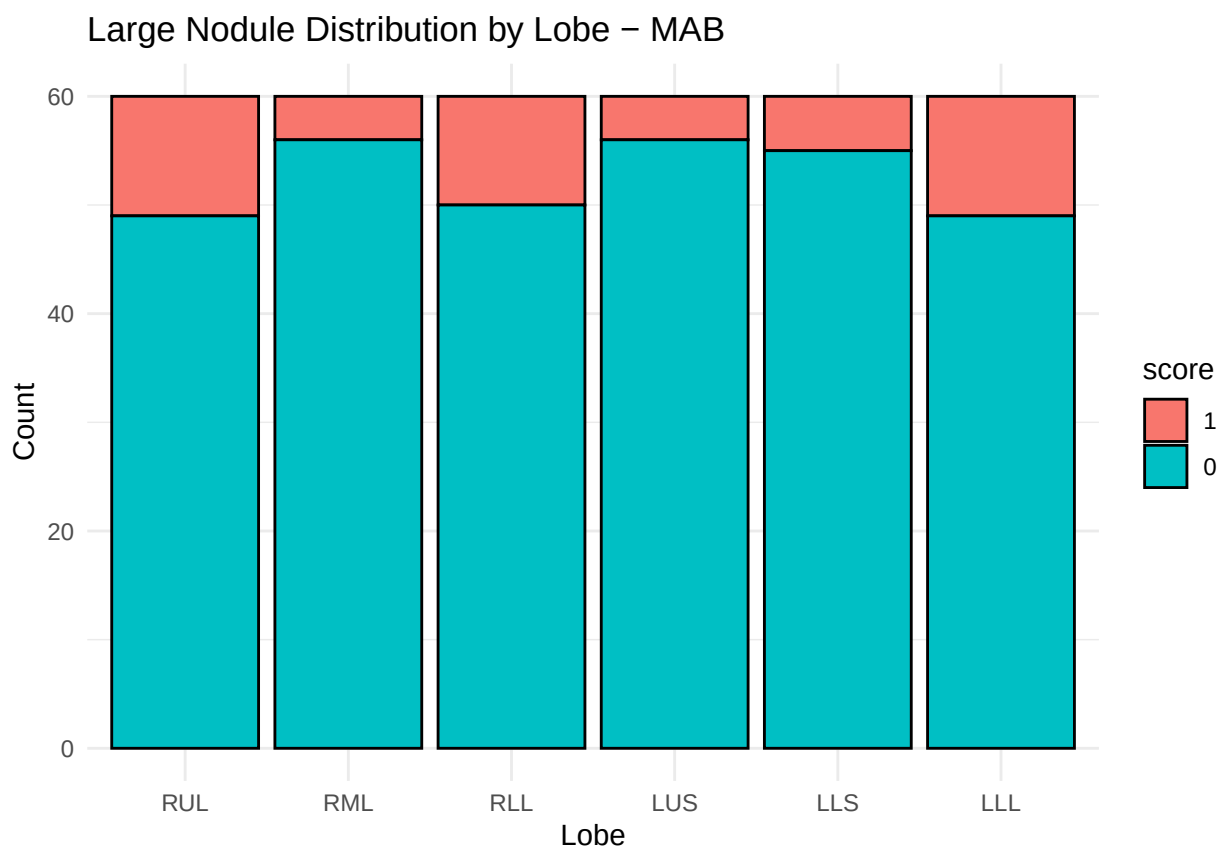


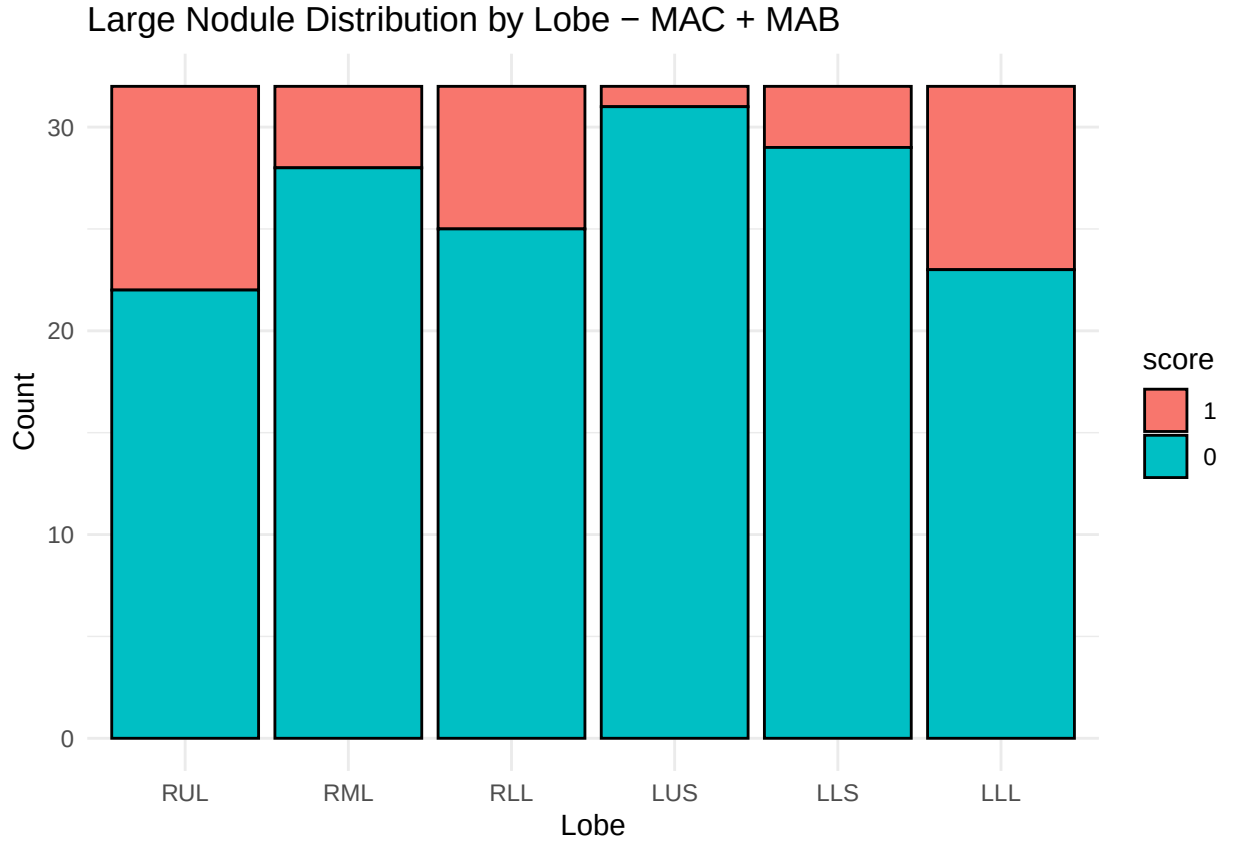
Large Nodules

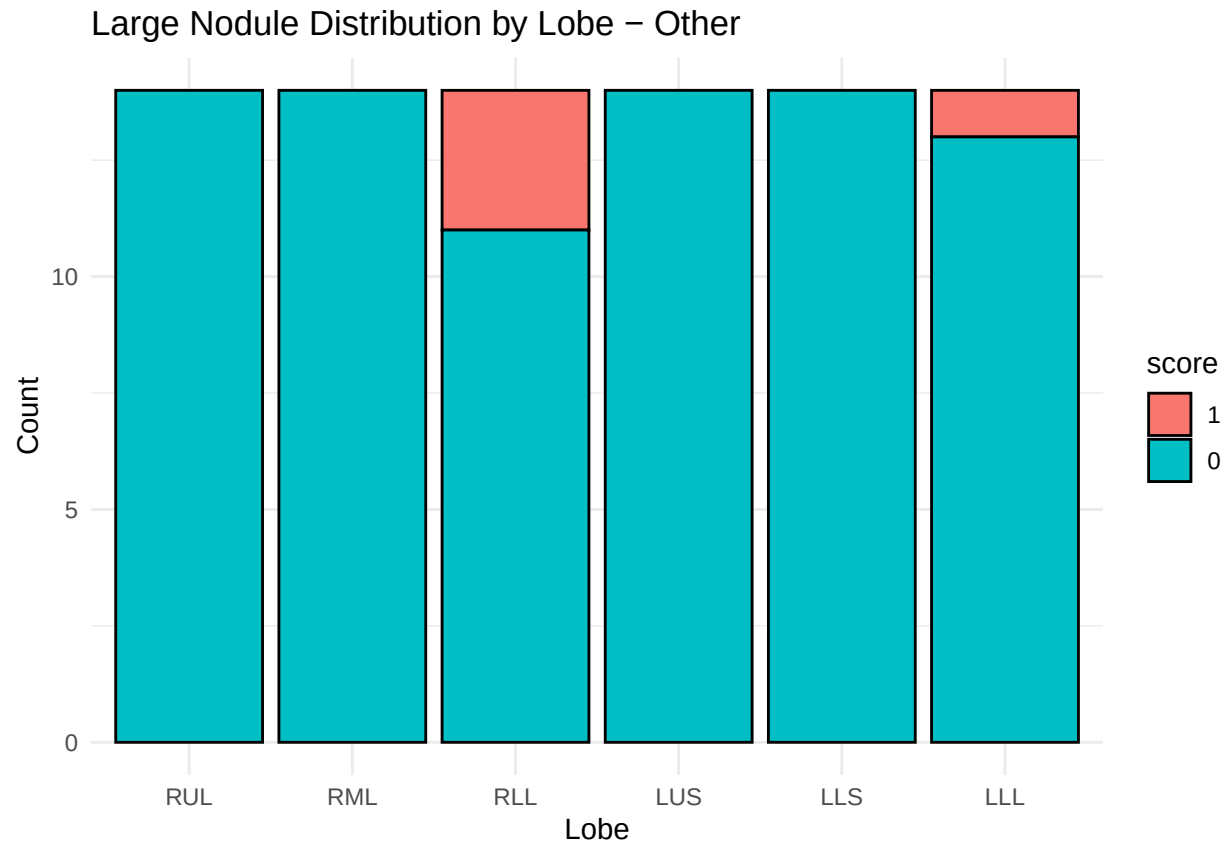
```
plots <- ln %>%
  mutate(
    score = factor(score, levels = c(1,0)),
    lobe = factor(lobe, levels = c("RUL","RML","RLL","LUS","LLS","LLL")),
    pathcode = factor(pathcode, levels = 1:4,
                      labels = c("MAC","MAB","MAC + MAB","Other"))
  ) %>%
  split(.$pathcode) %>%
  lapply(function(df) {
    ggplot(df, aes(x = lobe, fill = score)) +
      geom_bar(position = "stack", color = "black") +
      labs(title = paste("Large Nodule Distribution by Lobe -", unique(df$pathcode)),
           x = "Lobe", y = "Count") +
      theme_minimal()
  })

# Print
plots[[1]]; plots[[2]]; plots[[3]]; plots[[4]]
```





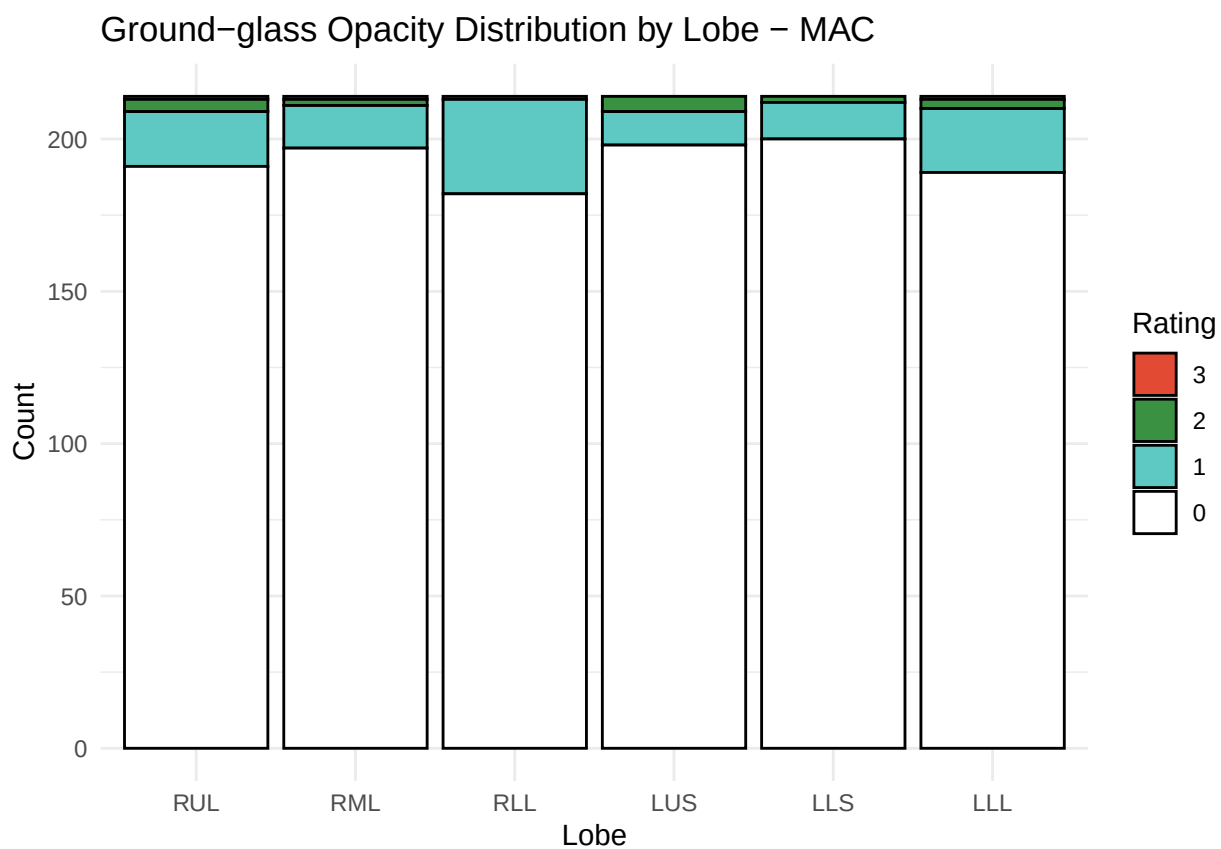


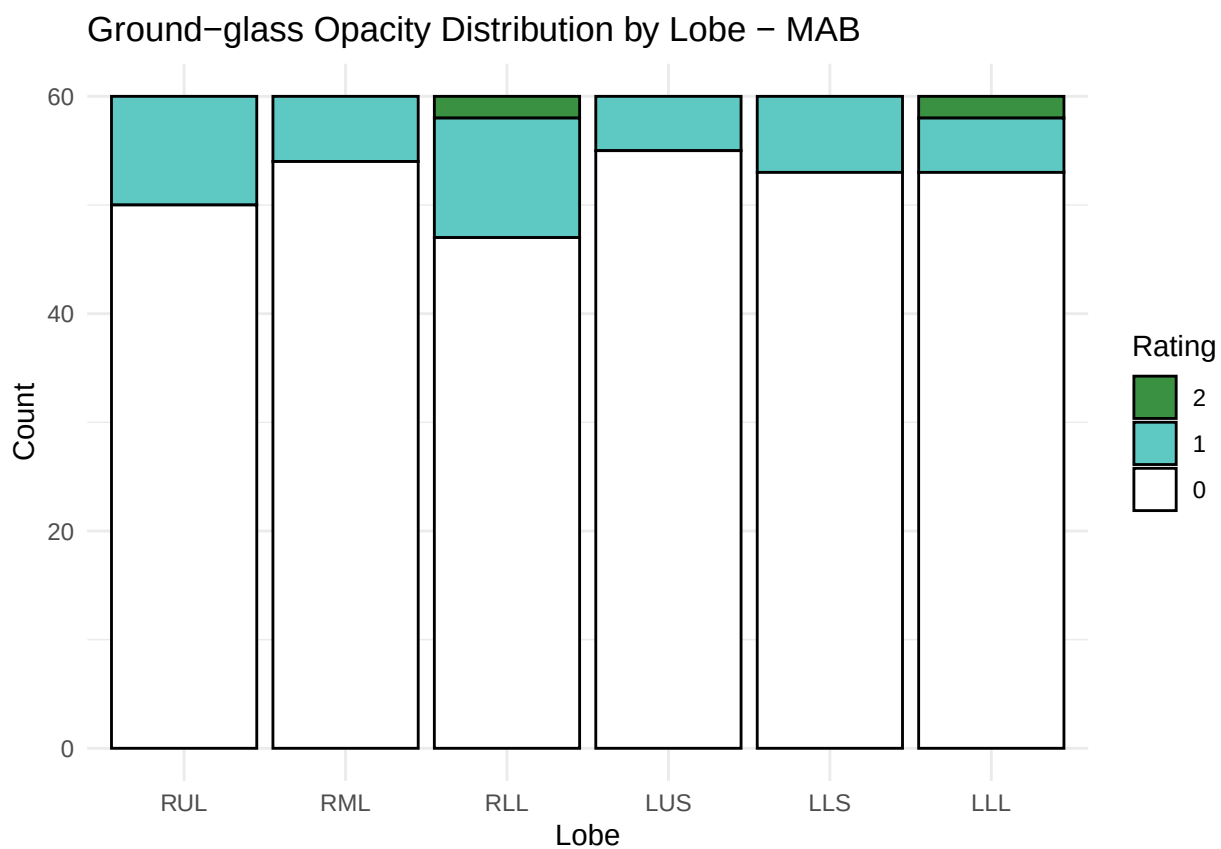


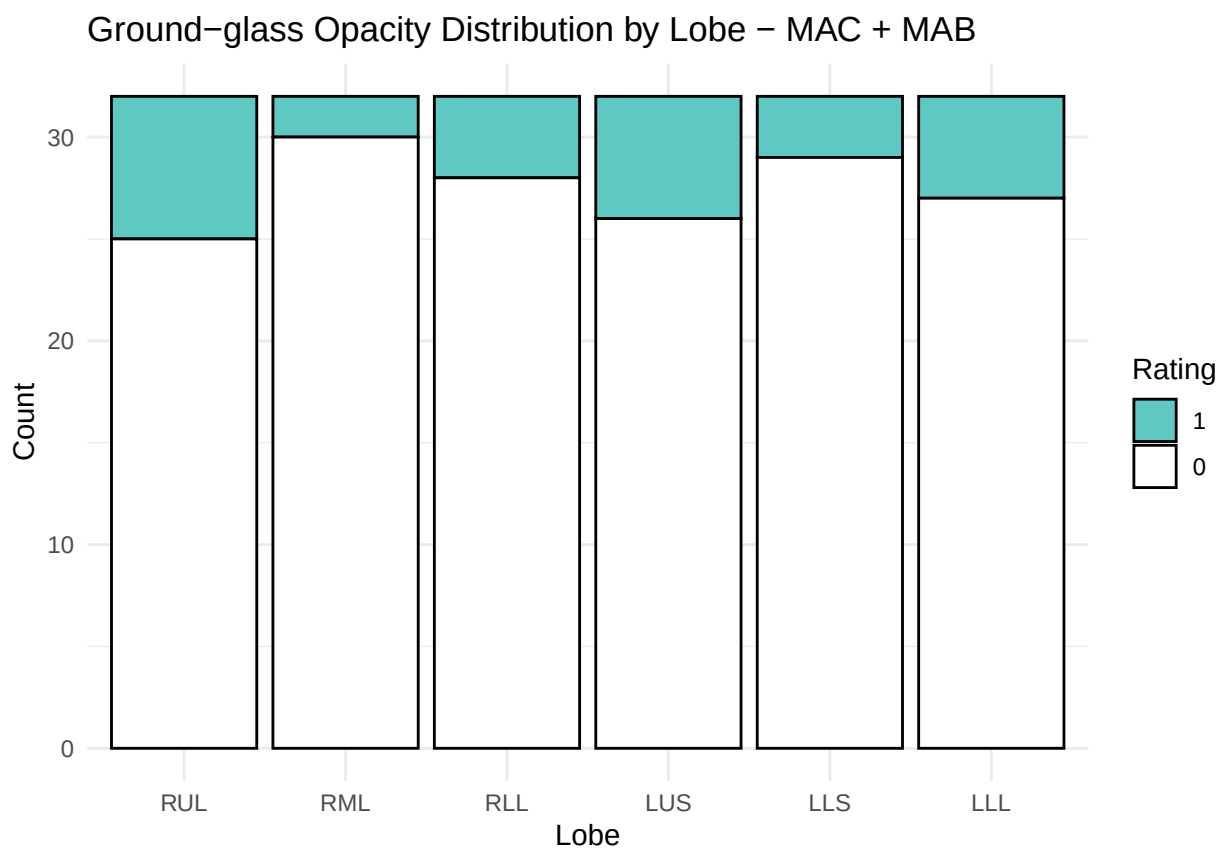
Ground-glass Opacities

```
plots <- ggo %>%
  mutate(
    score = factor(score, levels = c(3,2,1,0)),
    lobe = factor(lobe, levels = c("RUL","RML","RLL","LUS","LLS","LLL")),
    pathcode = factor(pathcode, levels = 1:4,
                      labels = c("MAC","MAB","MAC + MAB","Other"))
  ) %>%
  split(.$pathcode) %>%
  lapply(function(df) {
    ggplot(df, aes(x = lobe, fill = score)) +
      geom_bar(position = "stack", color = "black") +
      scale_fill_manual(values = score_colors_new, name = "Rating") +
      labs(title = paste("Ground-glass Opacity Distribution by Lobe -", unique(df$pathcode)),
           x = "Lobe", y = "Count") +
      theme_minimal()
  })

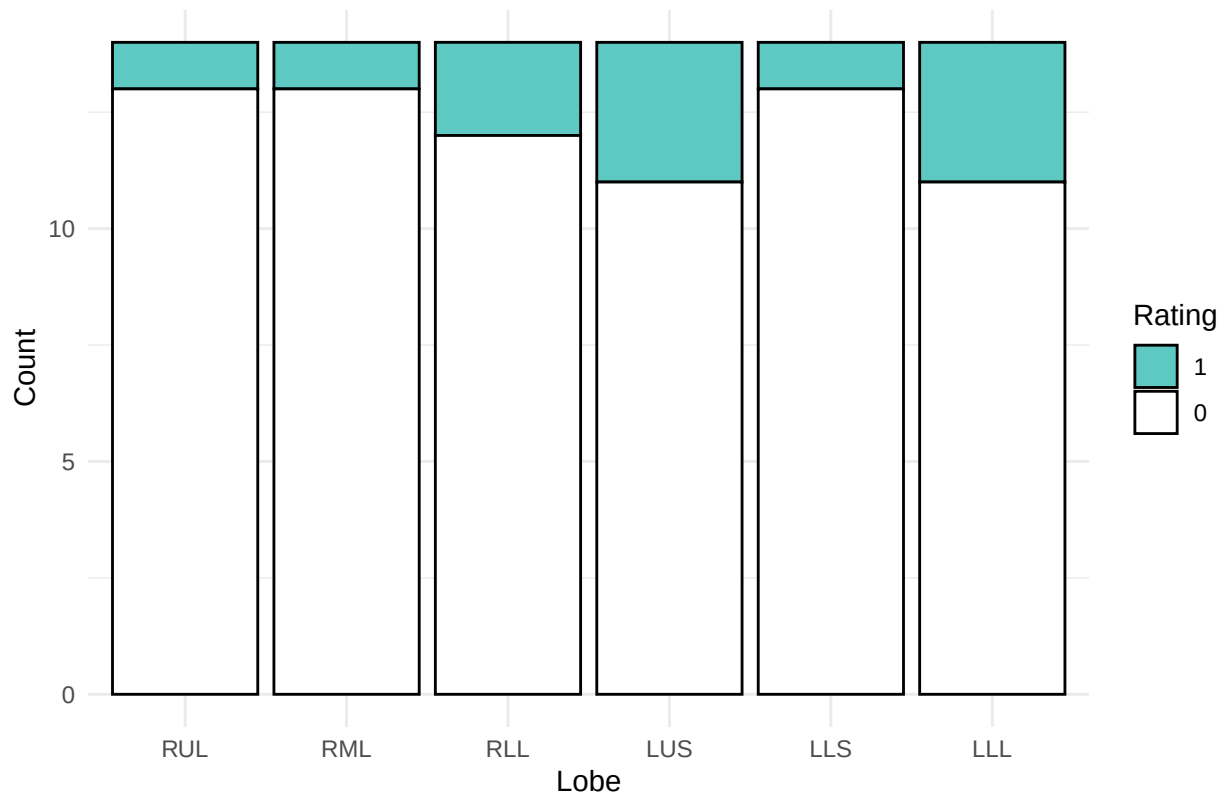
# Print
plots[[1]]; plots[[2]]; plots[[3]]; plots[[4]]
```







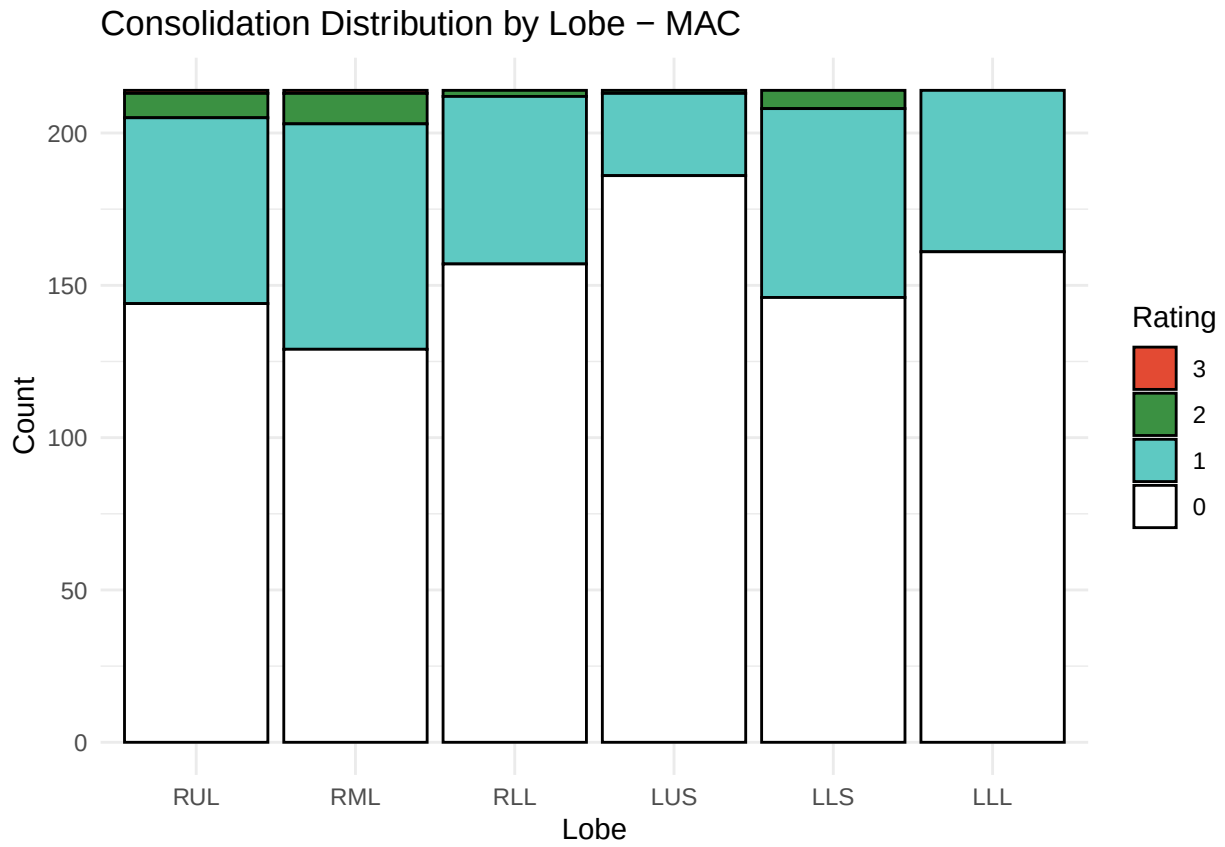
Ground-glass Opacity Distribution by Lobe – Other

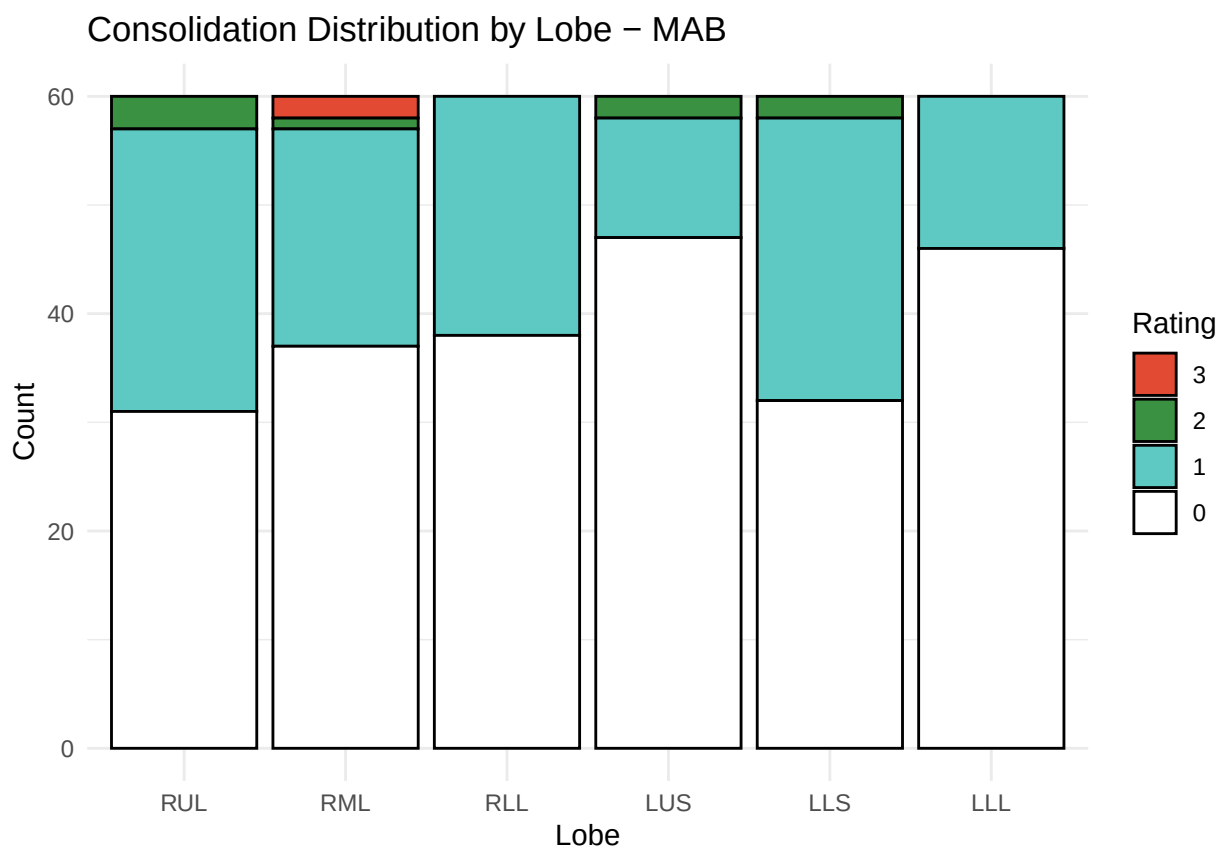


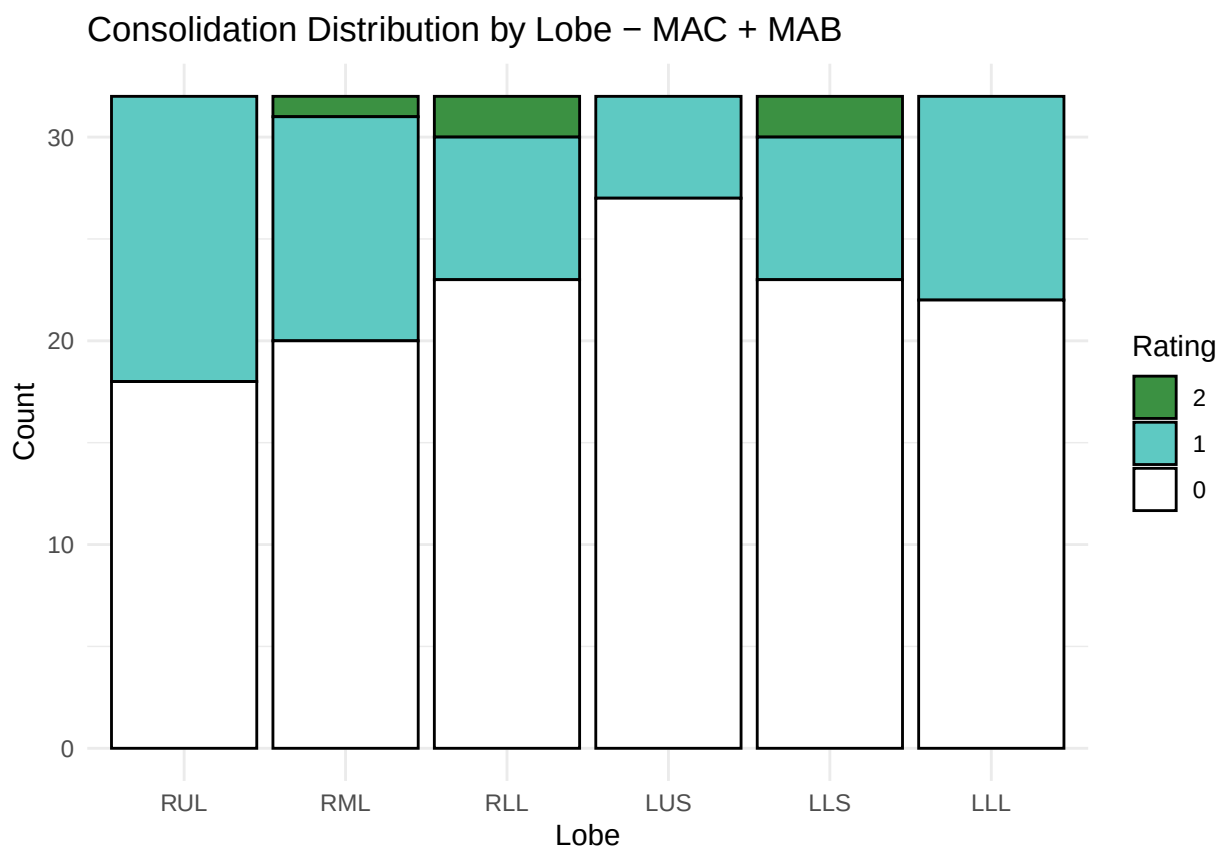
Consolidation

```
plots <- cons %>%
  mutate(
    score = factor(score, levels = c(3,2,1,0)),
    lobe = factor(lobe, levels = c("RUL", "RML", "RLL", "LUS", "LLS", "LLL")),
    pathcode = factor(pathcode, levels = 1:4,
                      labels = c("MAC", "MAB", "MAC + MAB", "Other"))
  ) %>%
  split(.$pathcode) %>%
  lapply(function(df) {
    ggplot(df, aes(x = lobe, fill = score)) +
      geom_bar(position = "stack", color = "black") +
      scale_fill_manual(values = score_colors_new, name = "Rating") +
      labs(title = paste("Consolidation Distribution by Lobe -", unique(df$pathcode)),
           x = "Lobe", y = "Count") +
      theme_minimal()
  })

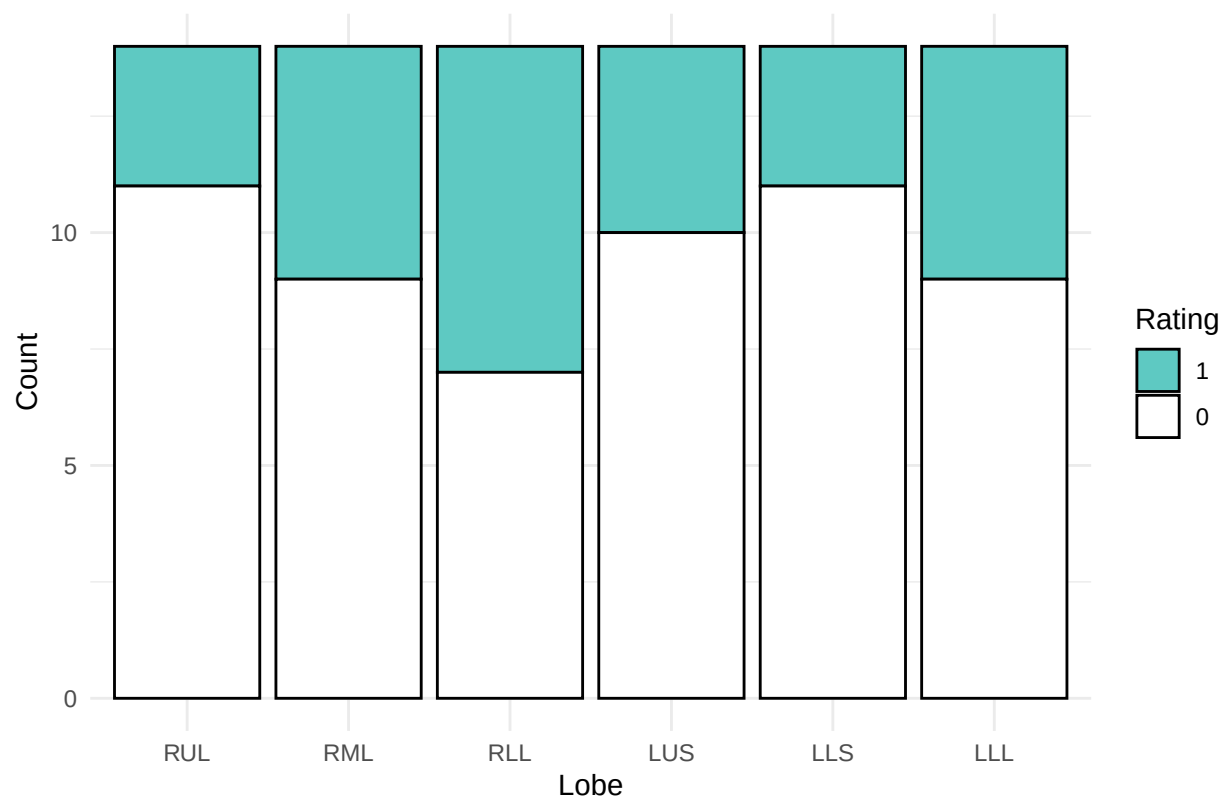
# Print
plots[[1]]; plots[[2]]; plots[[3]]; plots[[4]]
```







Consolidation Distribution by Lobe – Other

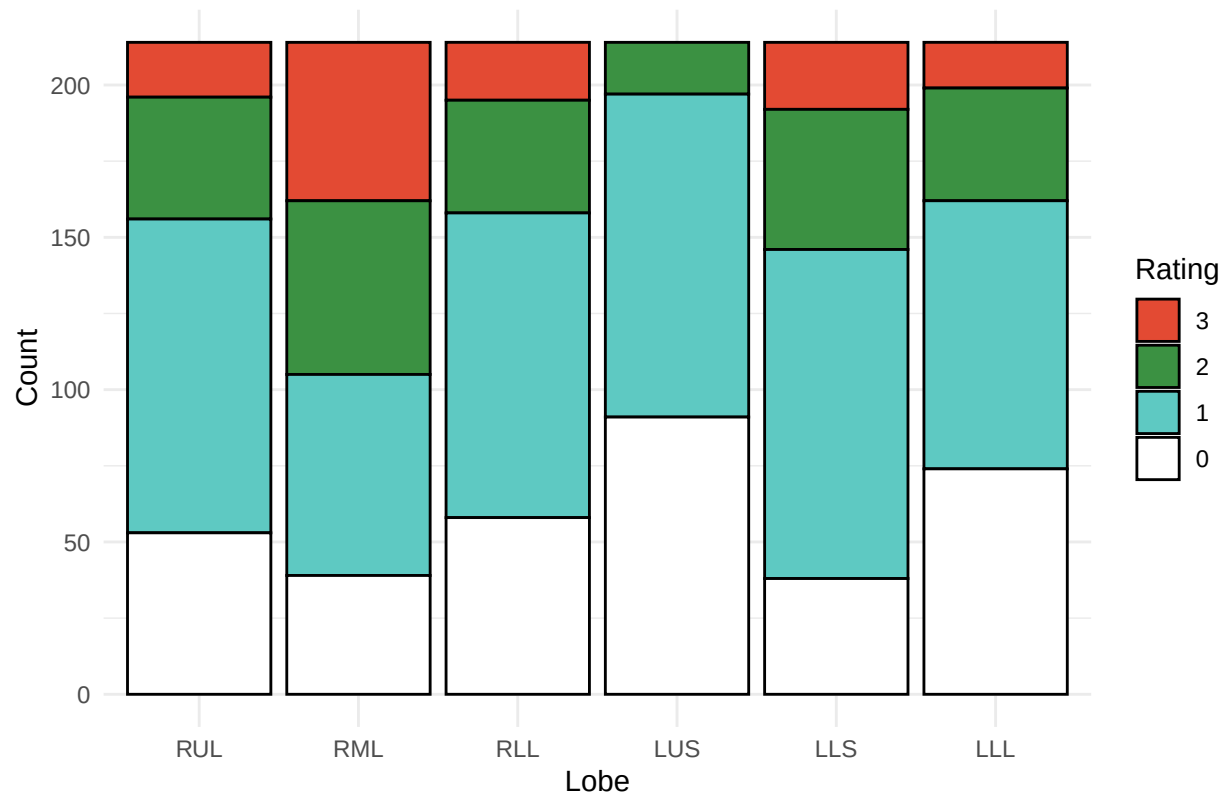


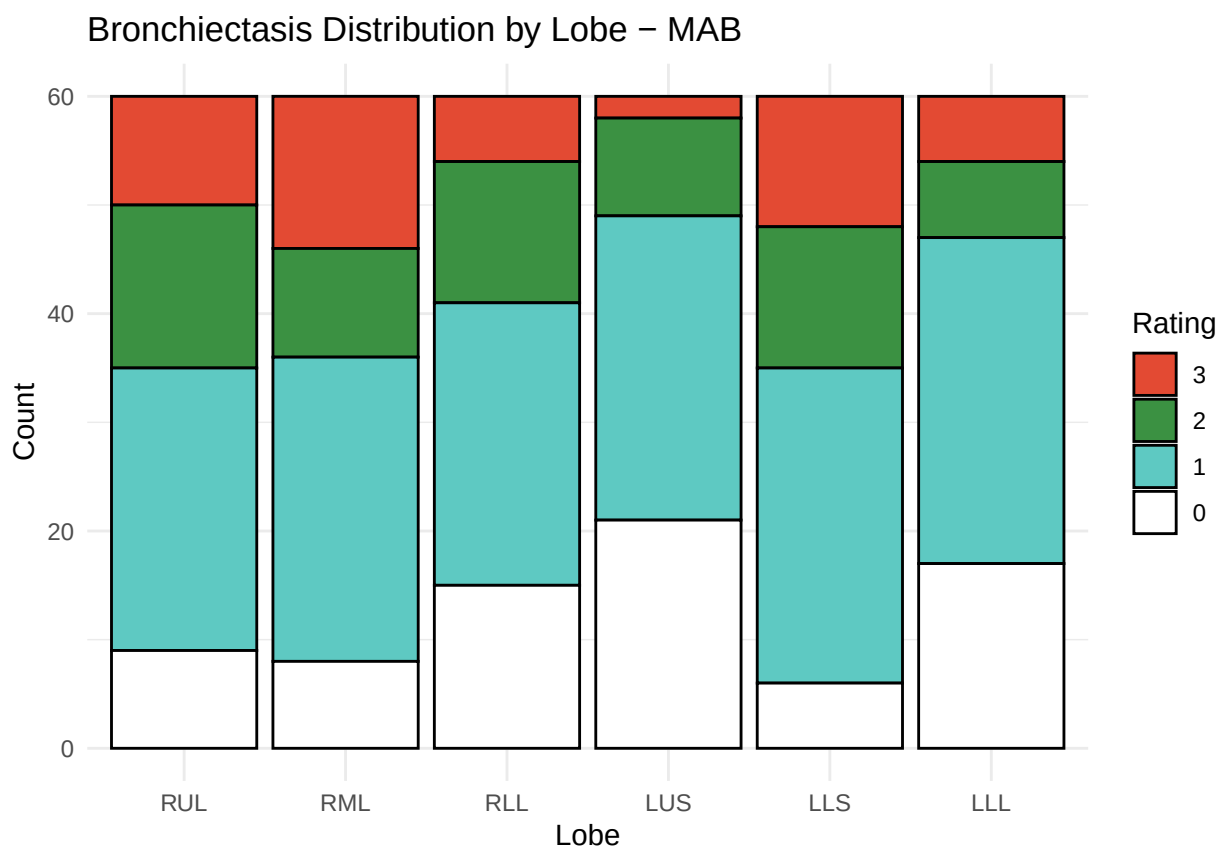
Bronchiectasis

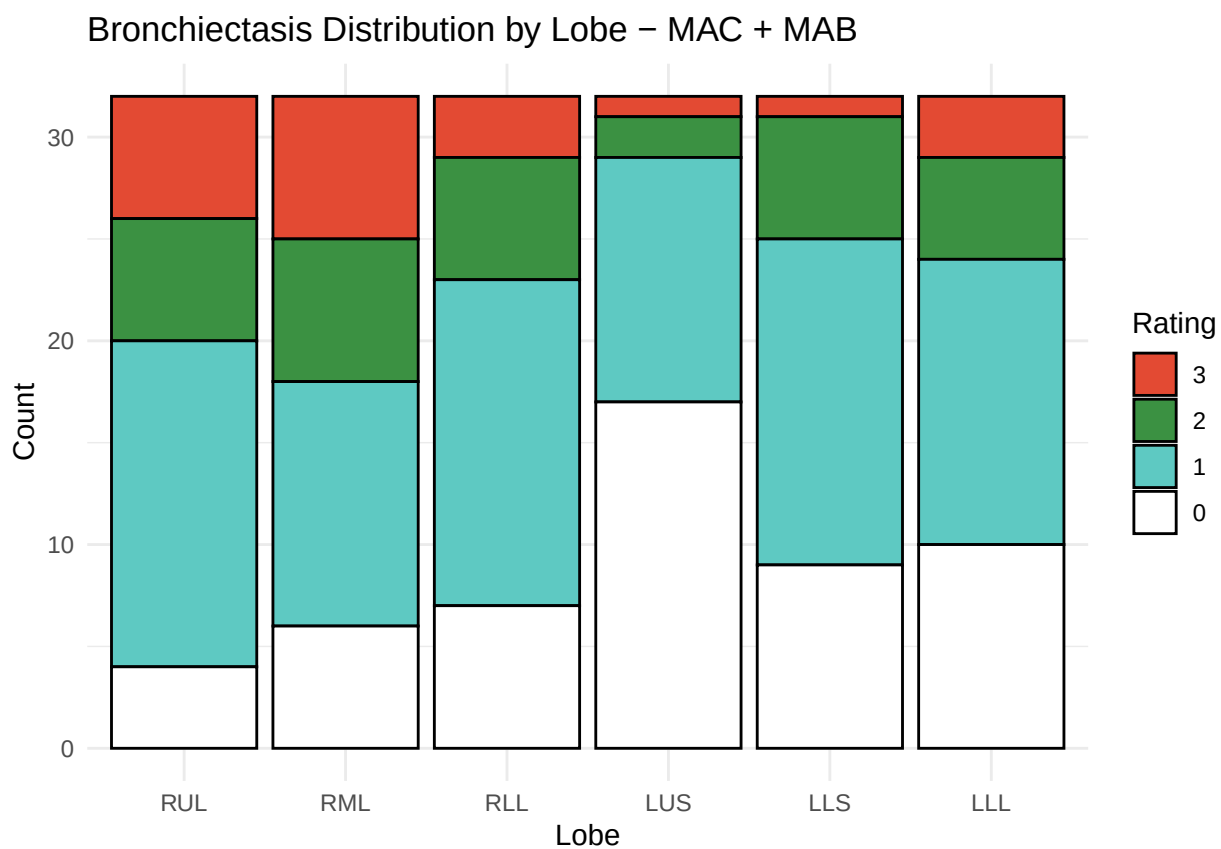
```
plots <- branch %>%
  mutate(
    score = factor(score, levels = c(3,2,1,0)),
    lobe = factor(lobe, levels = c("RUL", "RML", "RLL", "LUS", "LLS", "LLL")),
    pathcode = factor(pathcode, levels = 1:4,
                      labels = c("MAC", "MAB", "MAC + MAB", "Other"))
  ) %>%
  split(.$pathcode) %>%
  lapply(function(df) {
    ggplot(df, aes(x = lobe, fill = score)) +
      geom_bar(position = "stack", color = "black") +
      scale_fill_manual(values = score_colors_new, name = "Rating") +
      labs(title = paste("Bronchiectasis Distribution by Lobe -", unique(df$pathcode)),
           x = "Lobe", y = "Count") +
      theme_minimal()
  })

# Print
plots[[1]]; plots[[2]]; plots[[3]]; plots[[4]]
```

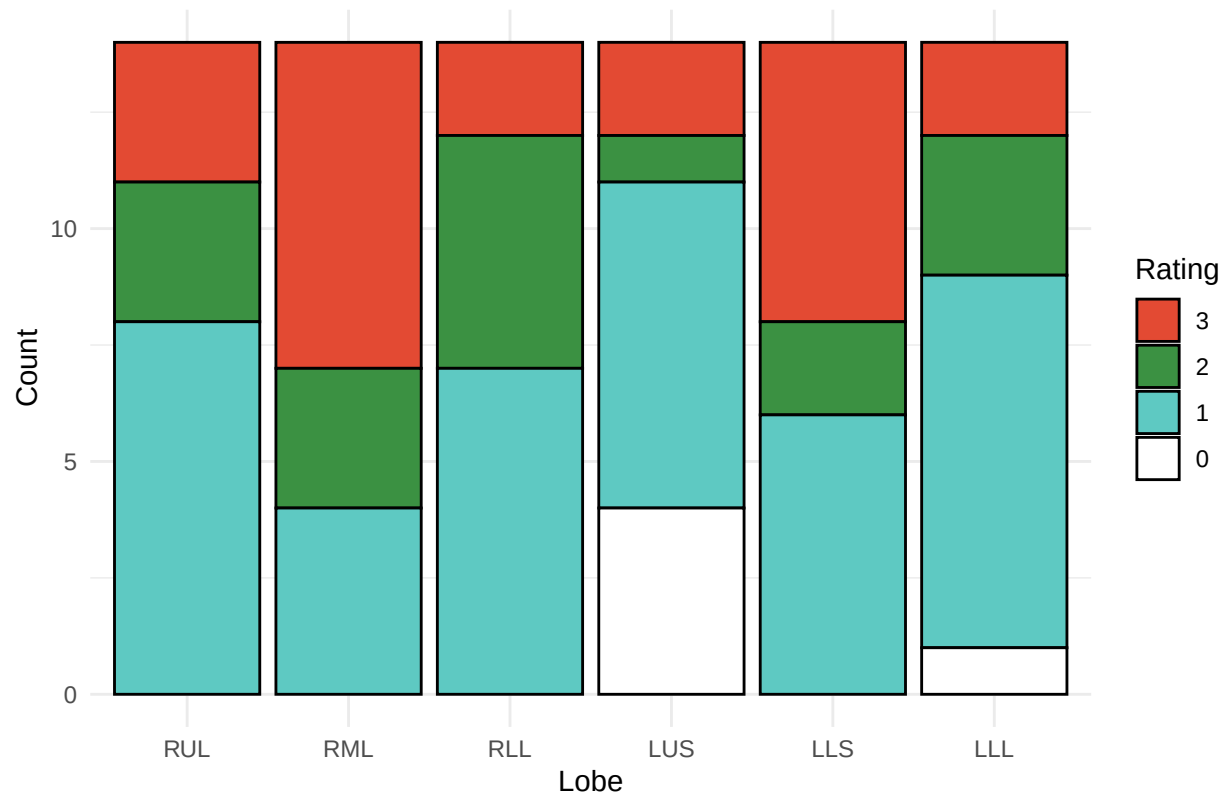
Bronchiectasis Distribution by Lobe – MAC







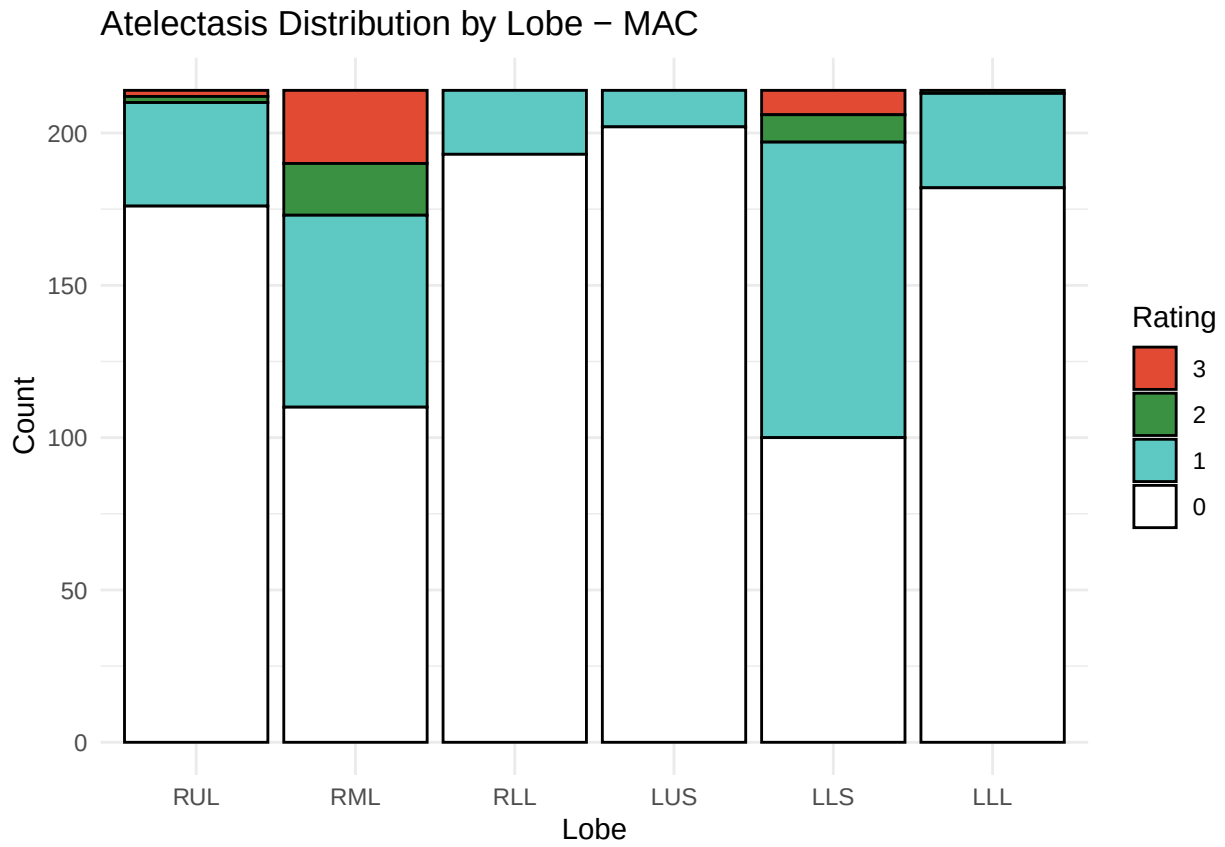
Bronchiectasis Distribution by Lobe – Other

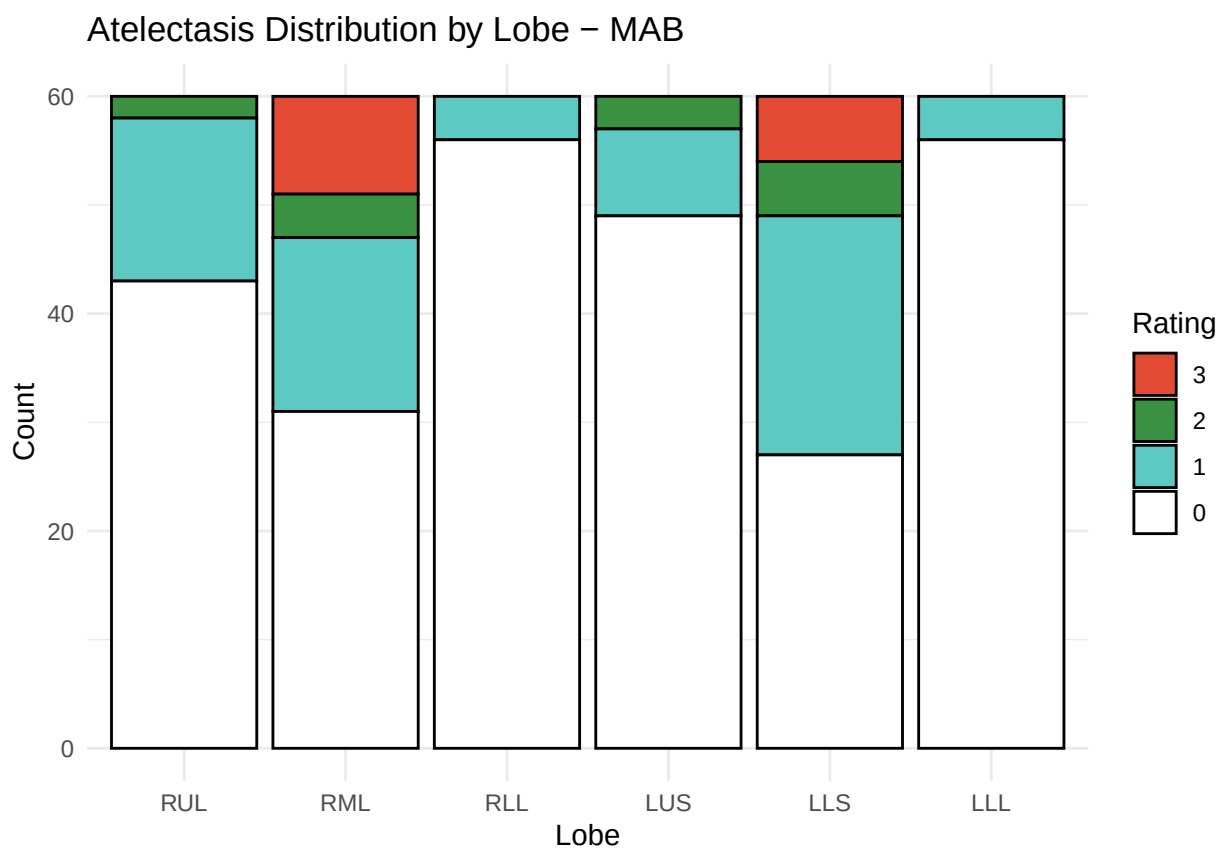


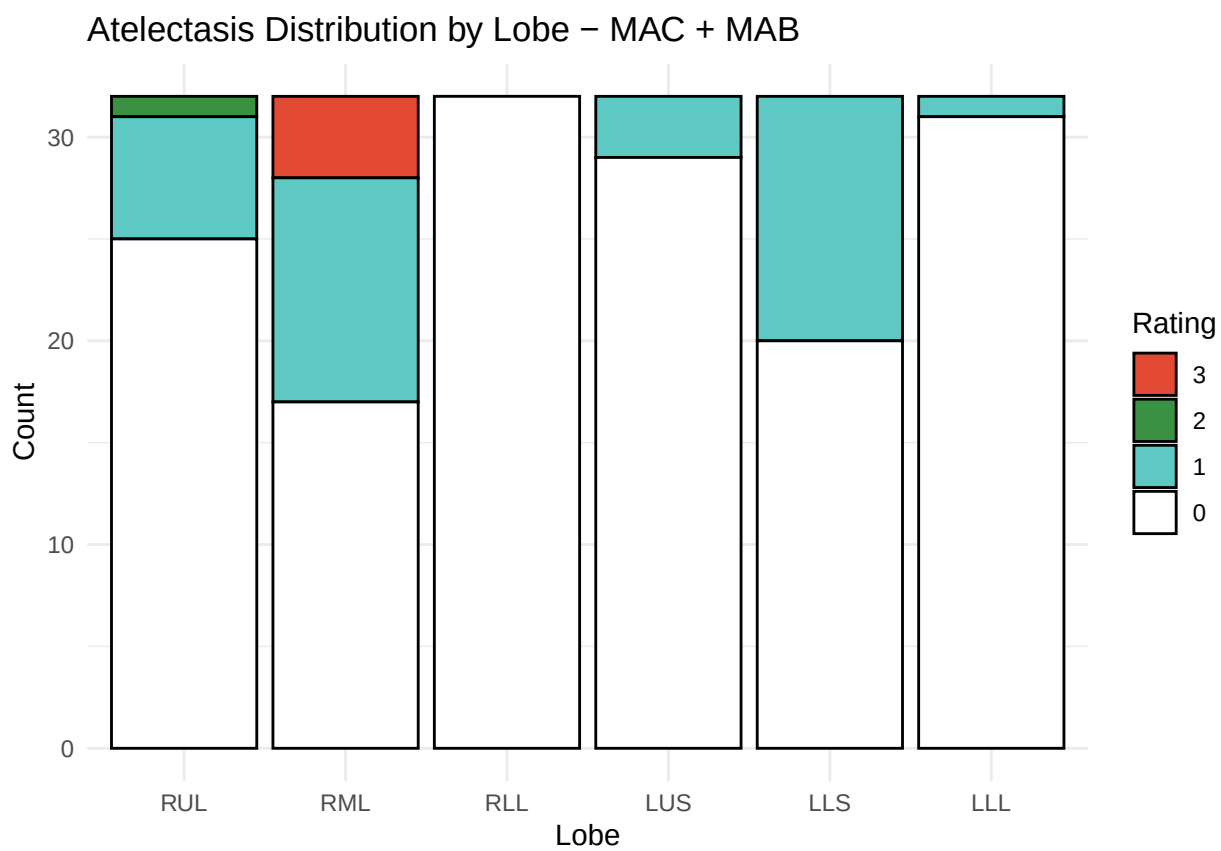
Atelectasis

```
plots <- atel %>%
  mutate(
    score = factor(score, levels = c(3,2,1,0)),
    lobe = factor(lobe, levels = c("RUL","RML","RLL","LUS","LLS","LLL")),
    pathcode = factor(pathcode, levels = 1:4,
                      labels = c("MAC","MAB","MAC + MAB","Other"))
  ) %>%
  split(.$pathcode) %>%
  lapply(function(df) {
    ggplot(df, aes(x = lobe, fill = score)) +
      geom_bar(position = "stack", color = "black") +
      scale_fill_manual(values = score_colors_new, name = "Rating") +
      labs(title = paste("Atelectasis Distribution by Lobe -", unique(df$pathcode)),
           x = "Lobe", y = "Count") +
      theme_minimal()
  })

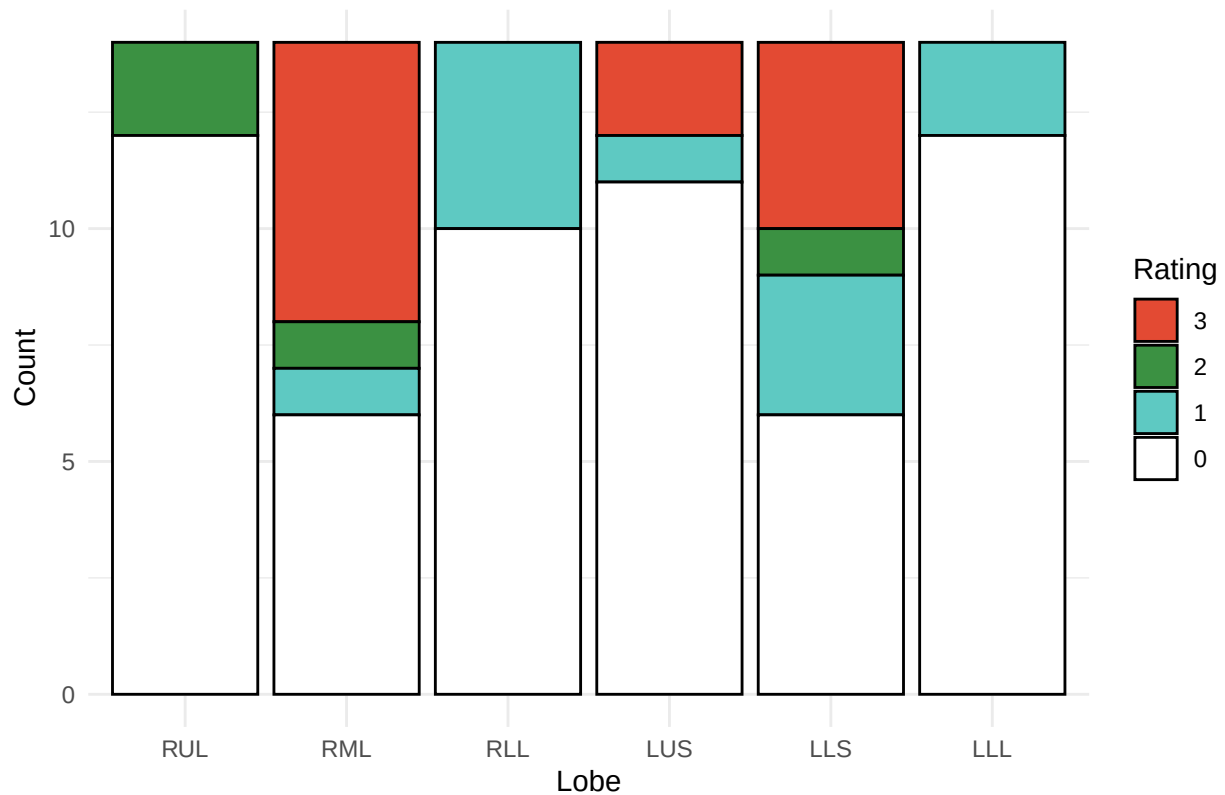
# Print
plots[[1]]; plots[[2]]; plots[[3]]; plots[[4]]
```







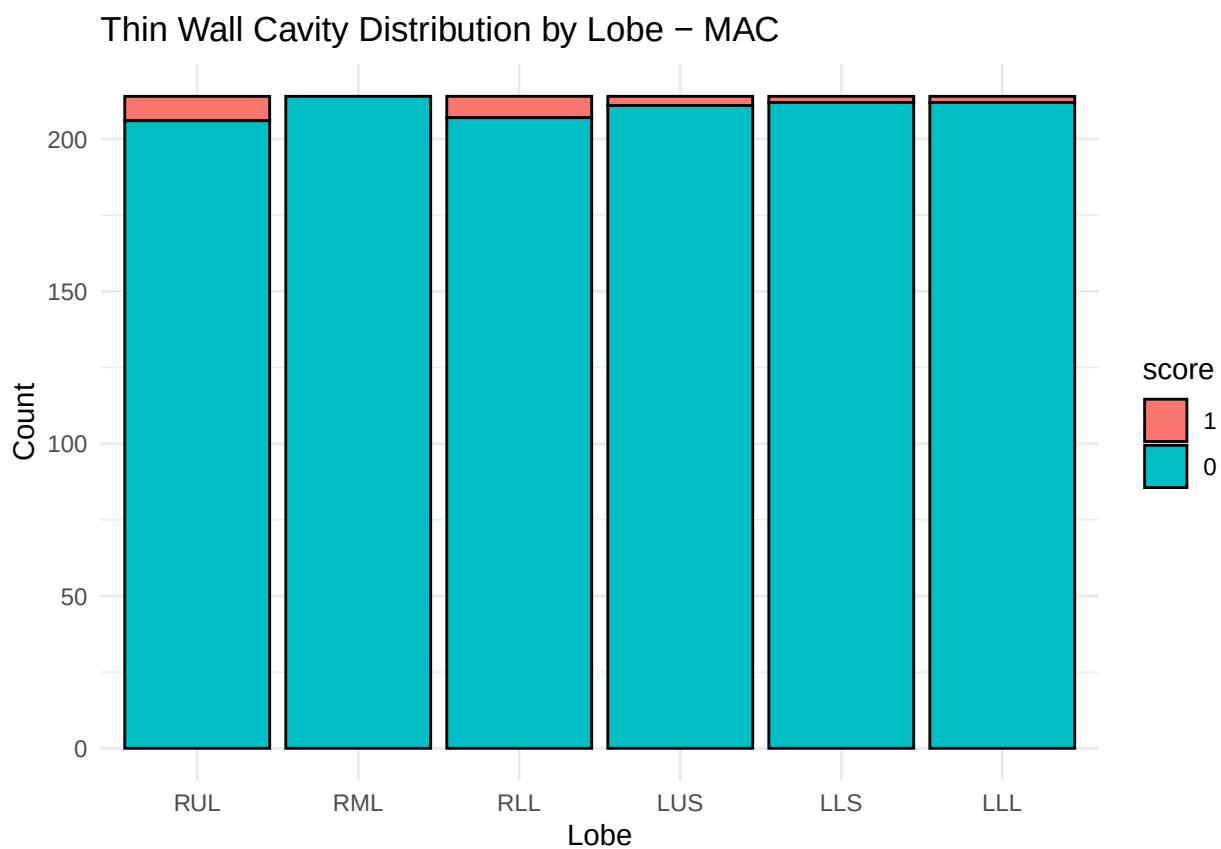
Atelectasis Distribution by Lobe – Other

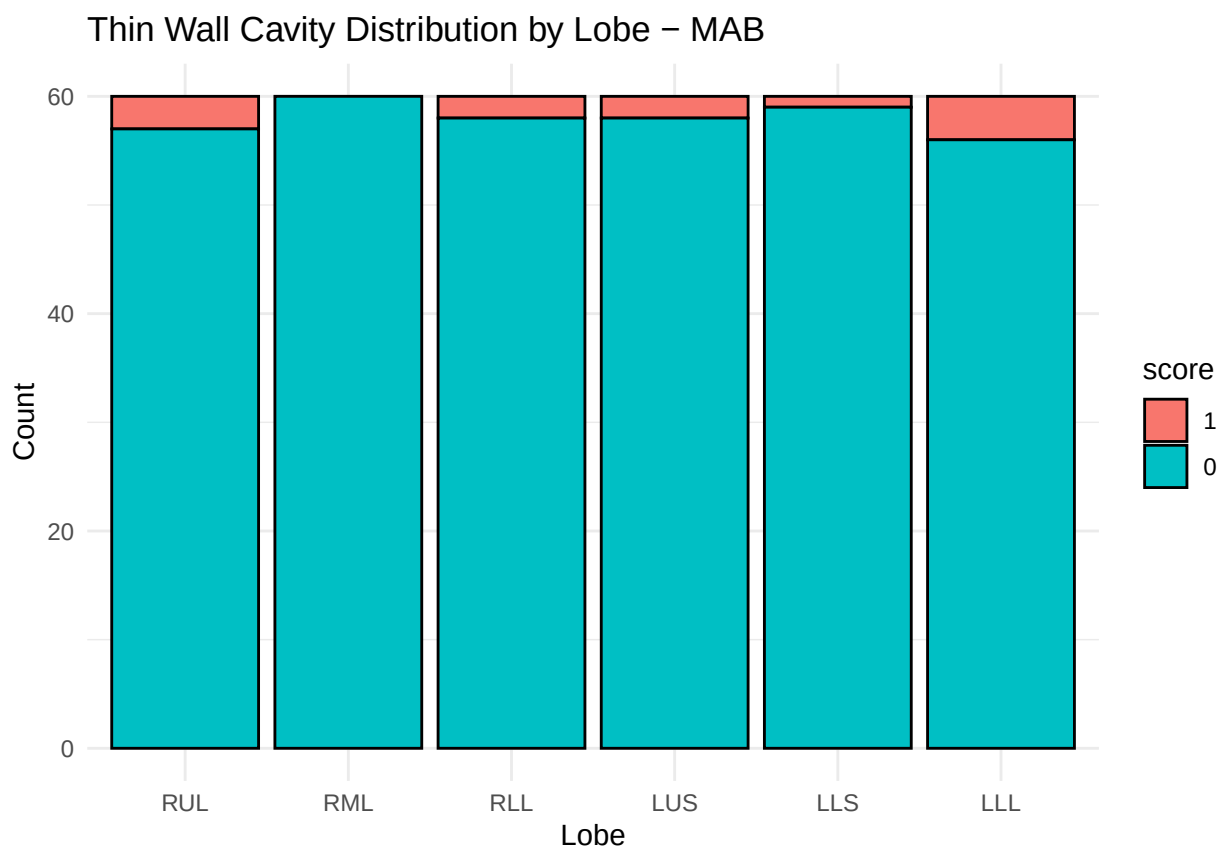


Thin Wall Cavity

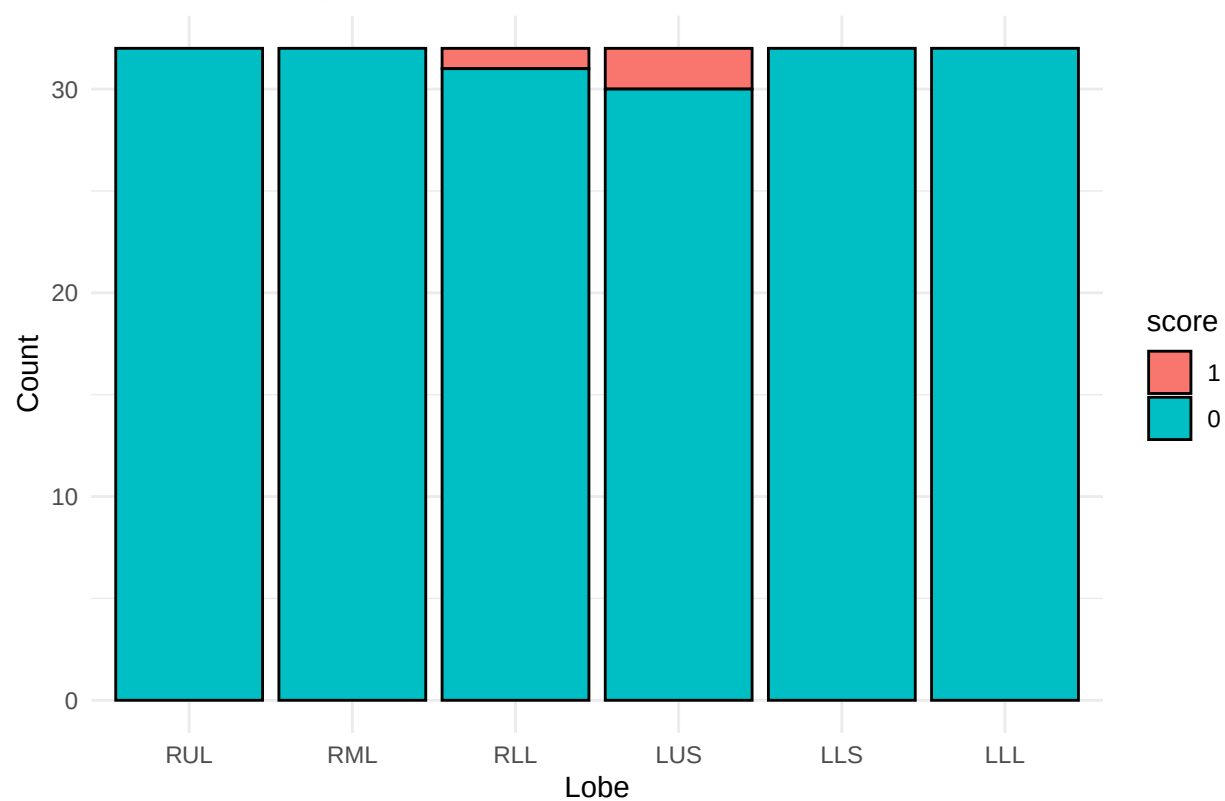
```
plots <- thin %>%
  mutate(
    score = factor(score, levels = c(1,0)),
    lobe = factor(lobe, levels = c("RUL","RML","RLL","LUS","LLS","LLL")),
    pathcode = factor(pathcode, levels = 1:4,
                      labels = c("MAC","MAB","MAC + MAB","Other"))
  ) %>%
  split(.$pathcode) %>%
  lapply(function(df) {
    ggplot(df, aes(x = lobe, fill = score)) +
      geom_bar(position = "stack", color = "black") +
      labs(title = paste("Thin Wall Cavity Distribution by Lobe -", unique(df$pathcode)),
           x = "Lobe", y = "Count") +
      theme_minimal()
  })

# Print
plots[[1]]; plots[[2]]; plots[[3]]; plots[[4]]
```

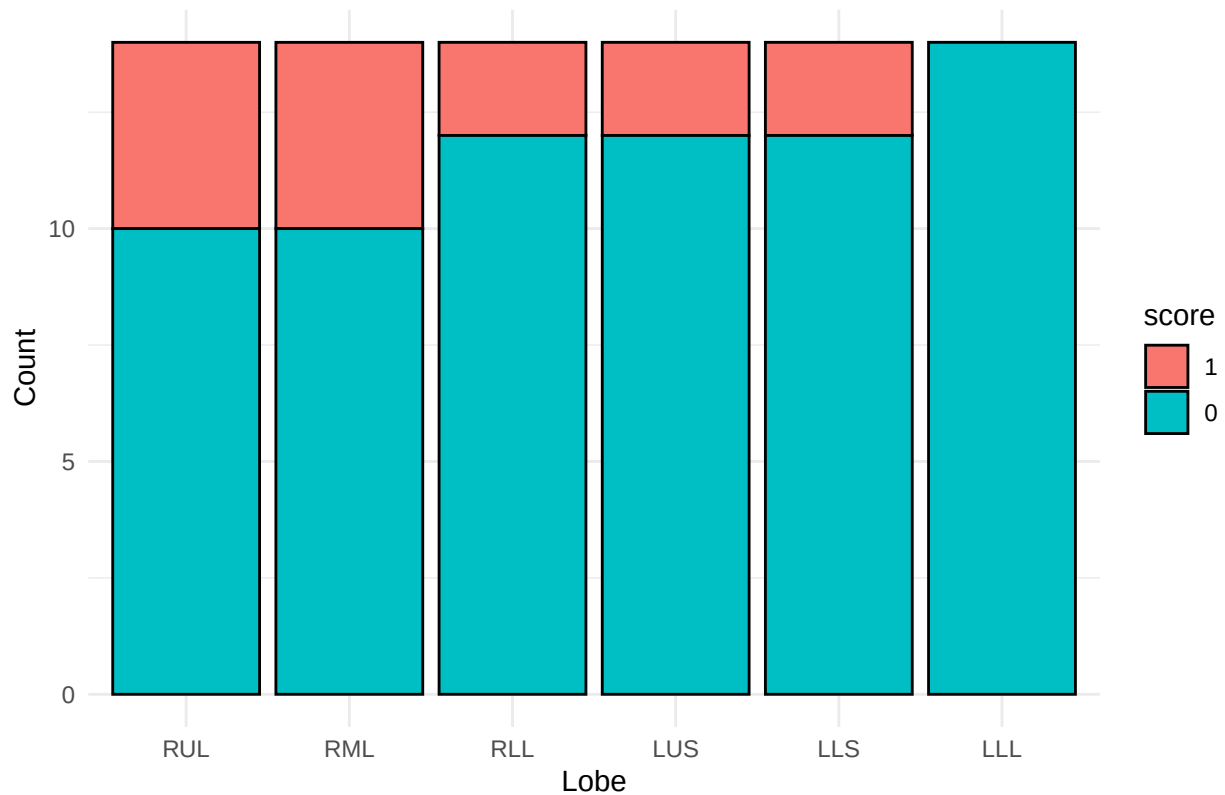




Thin Wall Cavity Distribution by Lobe – MAC + MAB



Thin Wall Cavity Distribution by Lobe – Other

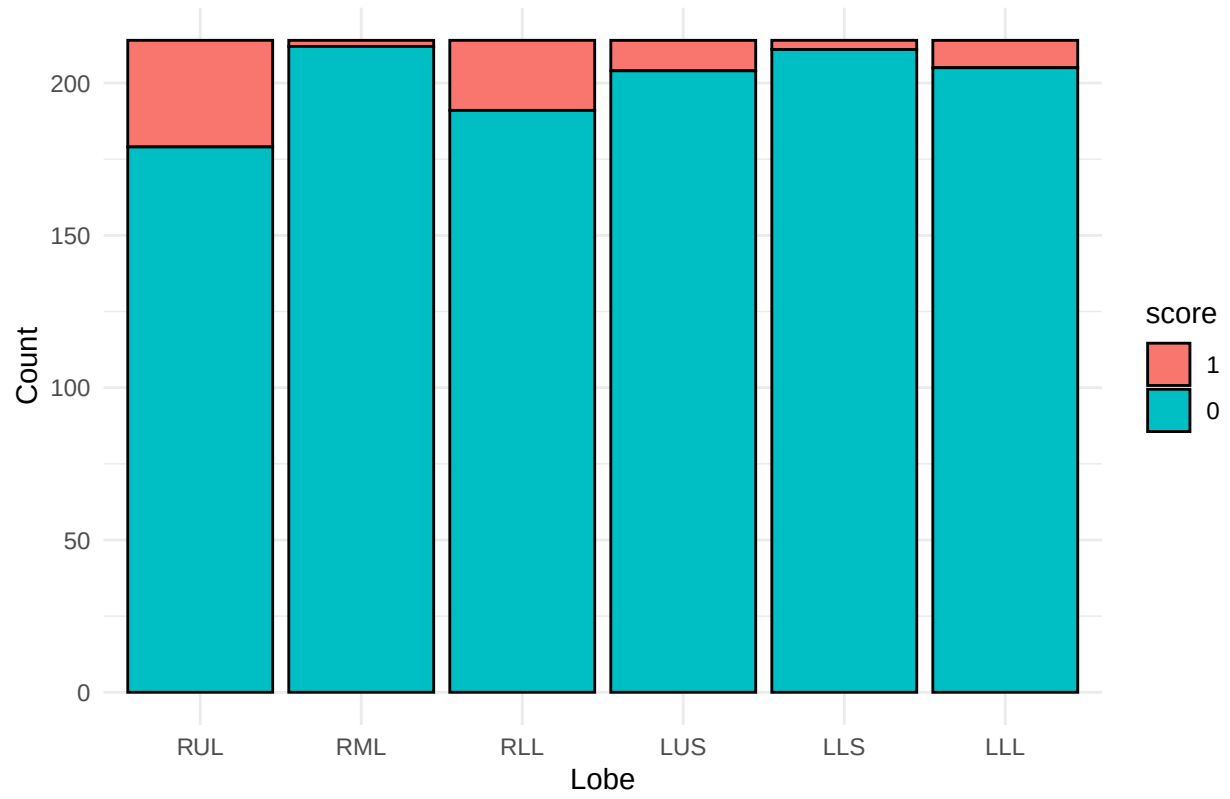


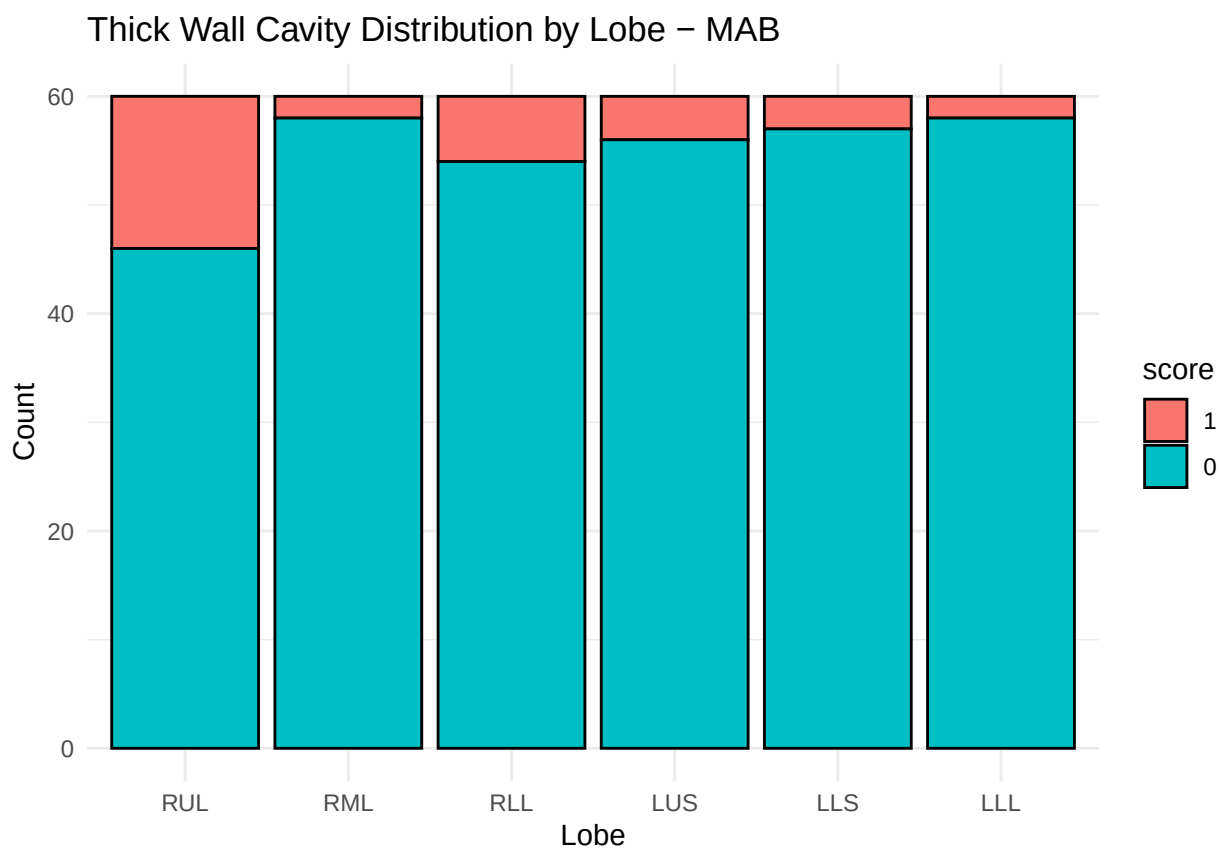
Thick Wall Cavity

```
plots <- thick %>%
  mutate(
    score = factor(score, levels = c(1,0)),
    lobe = factor(lobe, levels = c("RUL","RML","RLL","LUS","LLS","LLL")),
    pathcode = factor(pathcode, levels = 1:4,
                      labels = c("MAC","MAB","MAC + MAB","Other"))
  ) %>%
  split(.$pathcode) %>%
  lapply(function(df) {
    ggplot(df, aes(x = lobe, fill = score)) +
      geom_bar(position = "stack", color = "black") +
      labs(title = paste("Thick Wall Cavity Distribution by Lobe -", unique(df$pathcode)),
           x = "Lobe", y = "Count") +
      theme_minimal()
  })

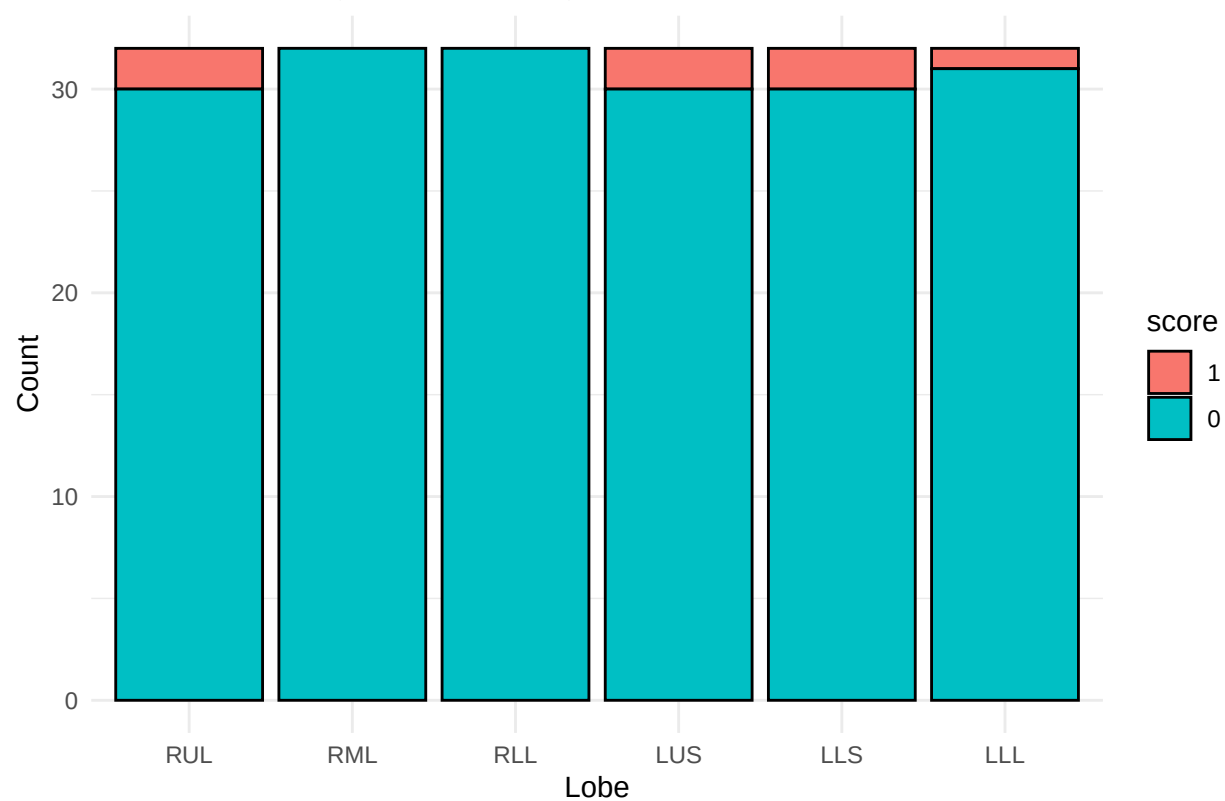
# Print
plots[[1]]; plots[[2]]; plots[[3]]; plots[[4]]
```

Thick Wall Cavity Distribution by Lobe – MAC





Thick Wall Cavity Distribution by Lobe – MAC + MAB



Thick Wall Cavity Distribution by Lobe – Other

